

PROJECT_INSTRUCTIONS.md — QB-TimeWarp

⚡ STEP 0: START MEMORY SERVICE (EVERY NEW SESSION!)

THIS MUST RUN BEFORE ANYTHING ELSE — NO EXCEPTIONS.

The MCP Memory Service holds all project knowledge, protocols, credentials, and mistake notes. Without it, you WILL forget critical rules.

Start the Server:

```
bash /home/ubuntu/mcp_memory_data/restart_memory_service.sh
```

Reconfigure the MCP Connection:

The preview URL changes every session. After starting the server:

```
# Get the new preview URL
echo $PREVIEW_URL
# The memory service runs on port 8765
# So the MCP URL will be: <PREVIEW_URL_with_-8765_in_subdomain>/mcp
# Example: https://abc123-8765.na103.preview.abacusai.app/mcp
```

Then use `perform_subtask_mcp_server_configuration` to reconnect with:

```
- server_name: mcp-memory-service
- server_url: <the new URL>/mcp
```

Verify It Works:

Use `memory_search` tool to query: “QB-TimeWarp protocols”

— You should get back notepad protocol, PowerShell enter, RDP resolution, etc.

What’s Stored in Memory:

- **Protocols:** Notepad = STOP, PowerShell needs {ENTER}, RDP = dynamic resolution
- **Credentials:** RDP, QB passwords, GitHub
- **Project State:** 7 fixes, commits, migration status
- **Workflow Rules:** One QB at a time, working copies only, GitHub deployment
- **Mistake Notes:** Past errors to avoid repeating

Persistence Details:

-  SQLite database PERSISTS on disk: `~/.local/share/mcp-memory/sqlite_vec.db`
 -  Server process does NOT persist — must restart each session
 -  MCP URL does NOT persist — must reconfigure each session
-

STEP 1: READ `/home/ubuntu/memory.md`

Read `/home/ubuntu/memory.md` for passwords, connection details, paths, current status, and the complete quick-start checklist.

ALSO READ THESE SUPPLEMENTAL FILES

Before starting work, check for and read these files if they exist:

Always Read

File	Purpose
<code>/home/ubuntu/memory.md</code>	 THE memory file — passwords, paths, status, next steps
<code>PROJECT_CONTEXT.md</code>	What the project does, design decisions, requirements
<code>CHANGE_LOG.md</code>	Full chronological history of every change made

Supplemental Files (ALL in `/home/ubuntu/`, NOT the project dir!)

```
# Run this EVERY session to read supplemental files:
cat /home/ubuntu/memory.md
cat /home/ubuntu/PROJECT_INSTRUCTIONS_ADDITION.md 2>/dev/null
cat /home/ubuntu/DEBUGGING_NOTES.md 2>/dev/null
cat /home/ubuntu/CHANGE_LOG_ADDITION.md 2>/dev/null
cat /home/ubuntu/DEBUGGING_SUMMARY_FOR_JOSEPH.md 2>/dev/null
cat /home/ubuntu/debugging_review_report.md 2>/dev/null
```

File	Location	Purpose
memory.md	/home/ubuntu/	THE canonical memory file — all credentials, paths, status
PRO-JECT_INSTRUCTIONS_ADDITION.md	/home/ubuntu/	Root cause analysis of import regression — 2 bugs, QBXML requirements, fix priorities
DEBUGGING_NOTES.md	/home/ubuntu/	Deep debugging: success detection bug, QBXML parsing errors, code architecture notes
CHANGE_LOG_ADDITION.md	/home/ubuntu/	Regression timeline: what broke in Turn 45, 359 phantom successes discovery
DEBUGGING_SUMMARY_FOR_JOSEPH.md	/home/ubuntu/	Concise debugging summary formatted for Joseph
debugging_review_report.md	/home/ubuntu/	Debugging review report

Legacy Files in Project Dir (still valid context)

File	Purpose
PROJECT_INSTRUCTIONS.MD (uppercase)	Original agent briefing from earlier session
DEBUGGING_NOTES.MD (uppercase)	Debugging notes copy in project dir

Reference As Needed

CURRENT IMPLEMENTATION STATUS (Updated 2026-05-19)

Critical Infrastructure - Schema Caching

Status:  ACTIVE (Commit `26e82b9`)

The QB 2021 schema is extracted **ONE-TIME ONLY** and cached at:

```
C:\QB-TimeWarp\Schemas\QB_Schema_QB_2021.json
```

Why: QB 2021's schema (supported fields, data types, XSD ordering) never changes. Once extracted, we load it from the cached file instead of opening QB 2021 every time.

Impact: Step 1 now loads instantly without requiring QB 2021 connection.

Critical Workflow - Sequential QB Operations

Status:  ENFORCED (Commit `a18d0da`)

RULE: Only ONE QuickBooks instance can be open at a time for SDK operations.

```
Step 1: Load Schema
   NO QB NEEDED (load from cached file)

Step 2: Export from QB 2023
   Open QB 2023
   Export all data
   CLOSE QB 2023 completely

Step 3: Transform Data
   NO QB NEEDED (pure data transformation)

Step 4: Import to QB 2021
   Open QB 2021
   Import all data
   CLOSE QB 2021 completely

Step 5: Validation
   Compare results, verify balances
```

What Changed:

- Pre-import validation (Step 3.5) DISABLED — it tried to open both QB 2023 AND QB 2021 simultaneously, violating the workflow
- Code commented out with detailed explanation in `Program.cs` lines 307-359

Critical Fix #7 - Timezone Offset Stripping

Status:  ACTIVE (Commit `1f18a89`)

Problem: QBXML does NOT support timezone offsets in date fields.

- QB 2023 exports: `2026-05-10T23:29:48-06:00` ❌
- QBXML requires: `2026-05-10T23:29:48` ✓

Solution: Strip all timezone suffixes (`-06:00` , `+05:30` , `Z`) from date/datetime fields before building QBXML.

Impact: Fixes 1,634 out of 2,072 previously failing records (TimeCreated, TimeModified, TxnDate, DueDate, etc.)

Implementation:

- Added `IsDateTimeField()` , `ContainsTimezoneOffset()` , `StripTimezoneOffset()` helper methods
 - Applied in `BuildFieldsXml()` and `BuildLineItemsXml()`
 - See `Services/DataImporter.cs` lines 2008-2012, 2108-2147
-

All 7 Critical Fixes - Summary

#	Fix	Commit	Status
1	Hierarchical name parsing	d0c9ffd	✓ ACTIVE
2	Force IsActive=true	d0c9ffd	✓ ACTIVE
3	Account type verification	d0c9ffd	✓ ACTIVE
4	Create missing reference types	d0c9ffd	✓ ACTIVE
5	Skip SDK 16.0-only fields	d0c9ffd	✓ ACTIVE
6	Success detection fix	d0c9ffd	✓ ACTIVE
7	Strip timezone offsets	1f18a89	✓ ACTIVE

GitHub Repository

URL: <https://github.com/LetsLearnAboutIt/QuickBooks-TimeWarp>

Latest Commits:

- a18d0da - Remove QB 2021 from export phase, enforce strict workflow
- 1f18a89 - FIX #7: Strip timezone offsets from QBXML dates
- 26e82b9 - Add schema caching (ONE-TIME extraction)
- d0c9ffd - Fixes 1-6 implementation

Workflow: All development happens on Linux (/home/ubuntu/QB-TimeWarp/), push to GitHub, pull on Windows VM (C:\QB-TimeWarp\)

File	Purpose
PATH_REFERENCE.md	All file paths + old→new rename mapping
ARCHITECTURE.md	Code structure, service layers, data flow
TESTING_CHECKLIST.md	Step-by-step testing procedures
POPUP_HANDLING.md	How to handle QuickBooks popups
SAFETY_FEATURES.md	Multi-layer protection system docs
SESSION_RESUME.md	Connection details (also in memory.md)
README_FOR_NEW_SESSIONS.md	Quick-start guide
PASSWORDS.txt	All credentials (gitignored)

PROJECT GOAL

Migrate data from **QuickBooks 2023** (SDK 16.0) to **QuickBooks 2021** (SDK 15.0) via a .NET 6.0 C# application that:

1. **Exports** all entity data from QB 2023 as JSON
2. **Transforms** the data for QB 2021 compatibility (field mapping, format preservation, SDK version adaptation)
3. **Imports** the transformed data into QB 2021 using staged, dependency-aware import
4. **Validates** the migration (entity counts, field comparison, financial reconciliation, format checks)

Key Constraints

- QBXML SDK is 32-bit only → build must target **x86**
- QuickBooks must be running with company file open for SDK to connect
- **Single-user mode** recommended for reliable SDK access
- Originals are NEVER modified — working copy system protects them

CRITICAL BUILD NOTES

```
# dotnet is NOT in PATH – always use full path:
/home/ubuntu/.dotnet/dotnet build

# Always build the PROJECT, never look for a .sln:
cd /home/ubuntu/QB-TimeWarp
/home/ubuntu/.dotnet/dotnet build    # uses QB-TimeWarp.csproj automatically

# Expected result: 0 Errors, 1 Warning (CS8602 in QBConnectionManager.cs:385)
```

WORKING WITH JOSEPH

- Joseph is a **40-year programmer** and MSP owner — he knows what he's talking about
- He watches the VM screen and programs alongside you — this is **collaborative**
- He wants **precise, complete documentation** with no shortcuts or placeholders
- **Don't barrel through** — pause at key stages and coordinate
- Keep QB windows **VISIBLE** — never minimize them during SDK operations
- Check `POPUP_HANDLING.md` for QuickBooks dialog handling procedures